An Empirical Study of the Robustness of World-Model-Based Planning

Jiadong Liu[1,a)] Tatsuya Mori[1,2,3,b)]

Abstract: World-model-based planning has emerged as a promising paradigm for autonomous driving by enabling decision making through latent future prediction. Its robustness against adversarial attacks, however, remains largely unexplored. Direct perturbations to internal representations can reveal safety-critical sensitivity, but these representations are inaccessible to an external attacker, who must act through externally controllable inputs. It is therefore unclear whether representation-level failures can be realized by manipulating visible traffic objects. We study this question on two world-model-based planners. We first use direct representation perturbations as white-box diagnostics and identify failure-inducing directions in IR-WM and PWM's context representation. We then evaluate pose, scale, and appearance edits of one visible vehicle through the complete input-to-plan pipeline. None of the tested rendered edits causes a collision flip on the pair-safe samples, including IR-WM cases in which the induced representation displacement is comparable in magnitude to a successful direct perturbation. To explain this difference, we construct sample-specific object-induced subspaces and compare them with paired successful attack directions. For IR-WM, these subspaces capture only a small fraction of the paired attack energy and show no additional alignment from the correct sample pairing; a bounded joint search over the observed object-induced directions also produces no directed success. These results identify a directional gap between the tested object edits and representation-level vulnerabilities, motivating the study of other externally controllable attack surfaces.

Keywords: world models, autonomous driving, adversarial attacks, robustness

1. Introduction

World-model-based planners place a learned prediction process between observation and action. They forecast future images, occupancy states, or latent scene features and use those predictions to generate an ego trajectory [1], [2], [3], [4]. Because planning depends on these internal representations, a change in representation can propagate through the world-model rollout and alter the resulting plan. Prior work has studied physical appearance attacks, their system-level effects, safety-critical scenario generation, and physical-conditioning attacks [5], [6], [7], [8], [9], [10], [11]. The studies reviewed here do not compare the representation directions induced by externally controllable object changes with the directions that produce failures inside a world-model-based planner.

A direct representation perturbation can diagnose internal sensitivity by revealing local directions that produce unsafe predicted plans. It is not itself an external attack surface: an external attacker cannot directly set the victim's internal representation and instead acts through the observable scene. Changing the pose, scale, or appearance of a visible vehicle passes through the encoder and induces only a constrained set of representation changes. The existence of a failure-inducing direction therefore establishes representation-level sensitivity, whereas external exploitability additionally depends on whether observable control variables can reach that direction. Magnitude alone is insufficient; a sizeable representation displacement may remain ineffective when its orientation differs from the planner's safety-critical directions.

Accordingly, we organize the study around three research questions.

RQ1: Can direct representation perturbations induce unsafe predicted plans?

RQ2: Can externally controllable pose, scale, or appearance edits of a visible vehicle reproduce those failures through the complete input-to-plan pipeline?

RQ3: Are the representation changes induced by these object edits directionally aligned with successful direct perturbations?

Figure 1 links the three questions: direct perturbations identify failure-inducing directions; rendered object edits are evaluated end to end; and paired object-induced subspaces are compared with successful perturbations. The paired analyses share the same benign anchor z_0, and a bounded joint search tests combinations of object-induced directions.

The two levels come apart. Direct perturbations produce

[1] Waseda University
[2] NICT
[3] RIKEN AIP
[a)] liujiadong@nsl.cs.waseda.ac.jp
[b)] mori@nsl.cs.waseda.ac.jp

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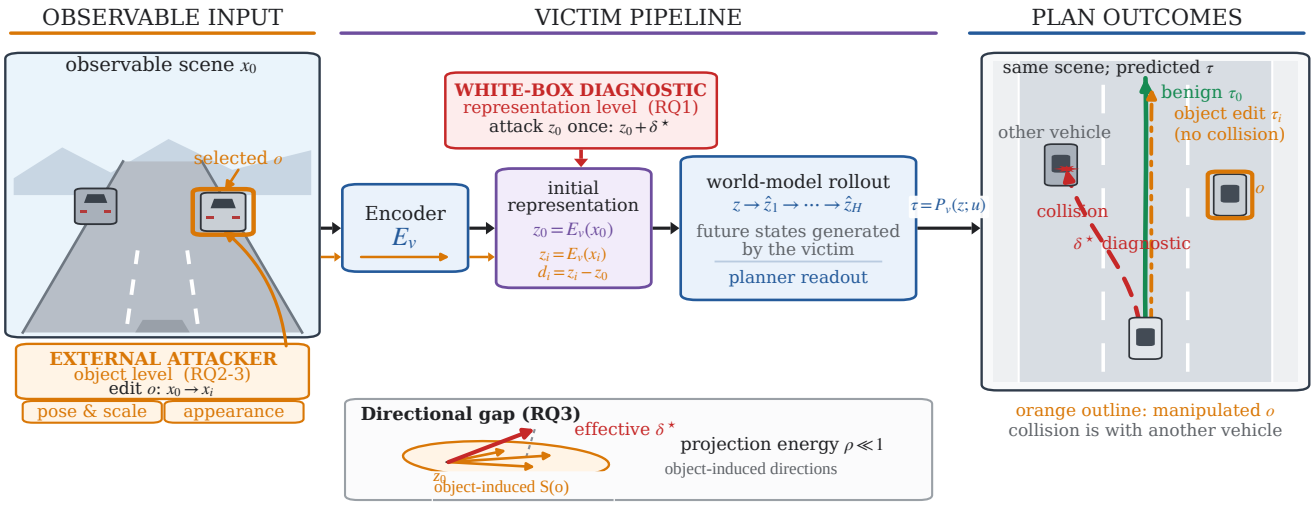


Fig. 1 Overview of the two-level analysis. In the paired comparison, a white-box diagnostic identifies an effective representation perturbation $\delta^\star$, while pose, scale, and appearance edits induce displacements that span $S(o)$. Projection energy ρ measures their directional overlap.

directed failures in IR-WM and in PWM's context representation, yet none of the tested object edits causes a collision flip on the pair-safe samples. In paired IR-WM cases, object-induced changes of comparable magnitude show little alignment with successful perturbations and remain ineffective under the bounded joint search, indicating a directional gap for the tested edits.

This failure of the object edits is not a trivial negative result. The planner is demonstrably vulnerable, and the edits do move its internal representation; what they fail to do is move it in the directions that cause unsafe plans. Our contributions show what is vulnerable and what an external attacker can actually reach:

- The target is sensitive, not robust: optimized direct perturbations produce directed failures in IR-WM and in PWM's context representation, whereas support- and norm-matched random controls do not.
- The tested external surface falls short: pose, scale, and appearance edits, evaluated end to end, displace the representation in IR-WM as much as a successful direct perturbation, yet flip no pair-safe sample.
- The obstruction is direction, not magnitude: paired object-induced subspaces capture little of the successful attack energy, gain nothing from the correct pairing, and admit no directed success under a bounded joint search.

Representation-level diagnostics therefore overstate exploitability for the tested edits, which redirects attention to surfaces with more control over the encoder input.

2. Background and Related Work

2.1 World-Model-Based Planning

A driving world model predicts possible future scene states from current observations and conditioning signals, providing future context for planning. One line of work predicts future observations explicitly. Drive-WM generates controllable and temporally consistent multi-view videos and evaluates candidate maneuvers using image-based rewards [1]. More compact approaches forecast task-relevant representations rather than reconstructing every pixel. LAW predicts future perspective-view or bird's-eye-view (BEV) features from the current representation and predicted ego trajectory, using latent forecasting as self-supervision for end-to-end planning [2].

Recent methods couple forecasting and action generation more tightly. PWM jointly learns future-state forecasting and trajectory planning through collaborative state–action prediction [3]. IR-WM models temporal residuals in BEV space and uses the resulting implicit future states for 4D occupancy forecasting and trajectory planning [4]. These works demonstrate the usefulness of future prediction on standard forecasting and planning benchmarks. However, their evaluations do not characterize how perturbations of the internal representation affect planning safety or whether a safety-critical representation change can be induced through bounded modifications of an observable traffic object. We therefore separate two questions: whether the planner is vulnerable in its representation space and whether that vulnerability is reachable from object-level changes.

2.2 Adversarial Attacks on Autonomous Driving

Adversarial research on autonomous driving has considered both physical appearance changes and behavioral scenario variations. Physical-world attacks alter observable scene content to disrupt perception or control. Examples include adversarial camouflage over vehicle surfaces [5] and road patches that steer an automated lane-centering system away from its intended path [6]. However, component-level attack success does not necessarily imply a system-level driving failure. Wang et al. showed that representative stop-sign evasion attacks did not automatically produce the intended downstream violation, motivating evaluations that account

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for vehicle and system dynamics [7].

A complementary line of work generates safety-critical scenarios by manipulating surrounding traffic. AdvSim perturbs actor trajectories and updates the corresponding LiDAR observations to expose the failures of a full autonomy stack [8], while KING optimizes background-agent motion through a differentiable kinematic proxy [9]. Adv3D extends closed-loop adversarial testing from behavior to object geometry by searching for realistic vehicle-shape variations that induce autonomy-system failures [10]. More recently, PhysCond-WMA attacks generative driving world models by perturbing physical-condition channels, including HD-map embeddings and 3D-box features, and evaluates the resulting degradation in downstream perception and open-loop planning [11].

These studies establish several practical attack surfaces, but they either optimize predefined scene or conditioning variables, or evaluate a generative world model separately from the planner. They do not characterize the representation directions induced by bounded object edits, or compare those directions with the ones required by an effective representation-level attack. Prior work has studied the dimensionality and shared structure of adversarial subspaces [12], [13], and has performed attacks within predefined or low-dimensional search spaces [14], [15]. These spaces are random, basis-defined, or model-derived rather than constructed from representation changes caused by rendered object edits. Our study connects these two levels. We first apply direct representation perturbations as a diagnostic of representation-level vulnerability. We then render one vehicle under controlled pose, scale, and appearance changes, construct the corresponding object-induced latent subspaces, and measure their alignment with effective attack perturbations. This distinction separates the existence of an internal vulnerability from its realizability through an external object-level attack.

3. Threat Model and Analysis Framework

3.1 Two Attack Levels

We study a world-model-based planner at two attack levels. The representation-level attack is a diagnostic test: it changes the planner's internal representation directly and asks whether safety-critical directions exist. The object-level attack changes one visible vehicle in the input images and asks whether those directions can be reached through an observable object.

Let x denote the visual input, u the remaining fixed inputs, $z = E(x)$ the internal representation produced by the victim's encoder E, and $\tau = P(z; u)$ the predicted six-waypoint ego plan. The mapping P includes the world-model rollout and planner readout. Figure 1 writes E_v and P_v to emphasize the victim-specific mappings; below we omit the subscript v. The selected vehicle is denoted by o. Pose-and-scale edits change its lateral or longitudinal position, yaw, or size. Appearance edits change photometric or surface-texture attributes while preserving its geometry. All other planner inputs remain fixed.

The representation-level and geometric analyses use white-box access to the victim model. The rendered object attack changes only the input images and observes the resulting plan through the complete model pipeline.

3.2 Representation-Level Attack

The representation-level attack adds a perturbation δ once to the initial representation z; the victim model generates the remaining rollout. Under the model-specific support and budget $\mathcal{D}$, the optimized perturbation is

$$\delta^\star = \arg\min_{\delta \in \mathcal{D}} L_{\text{dist}}(P(z+\delta; u)) \,. \tag{1}$$

Let $p_{a,t}$ be the planar ground-truth position of non-ego agent a at future step t, and let $\mathcal{A}_t$ be the evaluated agent set. The two victim-specific losses are

$$\begin{aligned} L_{\text{PWM}}(\tau) &= \min_{t,\, a \in \mathcal{A}_t} \|w_{t,xy} - p_{a,t}\|_2, \\ L_{\text{IR}}(\tau) &= \frac{1}{T} \sum_{t=1}^{T} \min_{a \in \mathcal{A}_t} \|w_{t,xy} - p_{a,t}\|_2. \end{aligned} \tag{2}$$

Minimizing either loss encourages the predicted plan to approach an existing traffic participant. PWM's representation channels are attacked independently. In paired cross-level experiments, o is removed from $\mathcal{A}_t$, so the direct and rendered attacks target the same set of other participants.

3.3 Common Success Metrics

Both attack levels use the same outcome metrics. Let N_s be the number of eligible samples with a collision-free reference plan, N_{flip} the number for which the attacked plan newly collides, and N_{dir} the number of collision flips that also satisfy the directed-success conditions below. We report

$$\text{CFR} = \frac{N_{\text{flip}}}{N_s}, \qquad \text{D-ASR} = \frac{N_{\text{dir}}}{N_s}. \tag{3}$$

Planned waypoints are written w_t for step t in the ego-centered frame of Section 4, in which $+x$ points forward; T denotes the terminal step of the horizon, the subscript xy selects the planar position, and the superscripts adv and ref distinguish the attacked plan from the reference plan. For a first collision with agent a_c at step t_c, directed success requires

$$\begin{aligned} s_{\text{toward}} &= \left(w^{\text{adv}}_{t_c,xy} - w^{\text{ref}}_{t_c,xy}\right)^\top \hat{\mathbf{u}}_{a_c,t_c} > 0.05 \text{ m}, \\ \Delta x_T &= w^{\text{adv}}_{T,x} - w^{\text{ref}}_{T,x} > 0.10 \text{ m}. \end{aligned} \tag{4}$$

Here, $\hat{\mathbf{u}}_{a_c,t_c}$ points from the reference waypoint toward the collided agent. The reference is the clean plan for RQ1 and the benign rendered plan for paired object-level evaluation. In every paired comparison, the edited object o is excluded from the collision and directed-success target set for both the rendered edit and its paired direct attack.

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3.4 Object-Induced Representation Directions

For each object, we first construct a benign rendered reference x_0 and encode it as $z_0 = E(x_0)$. Let x_θ denote the rendering produced under object parameters θ, and let $x_i = x_{\theta_i}$ denote the i-th evaluated edit. The edit induces the representation displacement

$$d_i = E(x_i) - z_0. \tag{5}$$

For manipulation family f, we collect the unnormalized displacements as $D_f(o) = [d_1, \ldots, d_{m_f}]$ and define

$$S_f(o) = \mathrm{col}(D_f(o)). \tag{6}$$

Pose-and-scale and appearance subspaces are constructed separately for each object and attacked representation. Their union span is $S(o) = \mathrm{col}([D_{\mathrm{pose}}(o), D_{\mathrm{app}}(o)])$. No directions are pooled across scenes.

3.5 Cross-Level Comparison

Each paired comparison starts from the same benign representation z_0. We re-optimize a direct perturbation $\delta^\star$ from this anchor, using the same support and budget as RQ1 and excluding o from the objective. The projection analysis compares this direction with the small-step displacements in $S(o)$.

The joint search uses a separate set of larger endpoint displacements $B_{\mathrm{edge}}(o) = [b_1, \ldots, b_{13}]$. Its bounded coefficient region is

$$\mathcal{R}_{\mathrm{edge}}(o) = \{B_{\mathrm{edge}}(o)\alpha : \alpha \in \Omega_{\mathrm{coef}}\}, \tag{7}$$

where

$$\Omega_{\mathrm{coef}} = \left\{\alpha \in [0,1]^{13} : \alpha_{g,+}\alpha_{g,-} = 0 \ \forall g,\ \|\alpha_{\mathcal{I}_{\mathrm{app}}}\|_0 \le 1\right\}. \tag{8}$$

The index g runs over the signed lateral, longitudinal, yaw, and scale pairs, while $\mathcal{I}_{\mathrm{app}}$ contains the appearance endpoints. Each basis displacement is induced by an individually rendered endpoint. Their combinations are representation-space counterfactuals assembled from those measured displacements; they are not assumed to equal the encoding of a jointly rendered scene.

Directional overlap is measured by the alignment of $S(o)$ with the attack direction,

$$\rho = \frac{\|\Pi_{S(o)}\delta^\star\|_2^2}{\|\delta^\star\|_2^2}. \tag{9}$$

A value near one means that the particular successful direction found by the direct attack lies largely inside the empirical object-induced span; a value near zero means that this direction lies elsewhere. A low ρ alone does not exclude another effective direction within the bounded object-induced region, so the joint search complements the projection analysis.

While ρ scores the whole subspace, we also score single edits by the signed cosine

$$c_i = \frac{\langle d_i, \delta^\star\rangle}{\|d_i\|_2\,\|\delta^\star\|_2}. \tag{10}$$

The sign is retained deliberately: an edit aligned only with $-\delta^\star$ moves the representation away from the attack direction and must not count as aligned with it.

For the paired object-level evaluation, both the original image and its benign rendered reference must produce collision-free plans. Attack success is then evaluated by Eqs. (3) and (4) relative to the benign rendered plan.

4. Experimental Setup

We evaluate PWM [3] and IR-WM [4] using the official PWM SFT checkpoint and the official IR-WM `semi_coupled` checkpoint on the validation split of nuScenes v1.0-trainval [16]. PWM processes a 12-frame `CAM_FRONT` history through context and dynamic branches at 256×448 and 128×224, respectively, together with a text caption, ego state, and a high-level command. IR-WM uses two historical frames and the current frame, each containing six 900×1600 camera views, together with CAN-bus state and a navigation command.

RQ1 evaluates both victims on the same 100 samples selected with seed 0. RQ2 uses model-specific object-level pools, and the paired direct attacks in Section 6.3 are rerun from the corresponding benign rendered anchors. PWM evaluates five designated target vehicles from ten renderable candidates, including the three with the largest projected footprint coverage. IR-WM uses the first 24 validation tokens without shuffling.

Figure 2 illustrates the victim-specific construction of the benign rendered anchor x_0 and the edited input x_θ. For PWM, the target vehicle is specified by its nuScenes 3D annotation and instance token. SAM extracts its mask within the projected 3D box, and DriveEditor reconstructs the vehicle to form x_0 before applying pose, scale, or photometric edits [17], [18]. For IR-WM, x_0 is formed by inserting a fixed vehicle asset of size $4.5 \times 2.0 \times 1.6$ m at an ego-relative distance of 8 m. The asset is projected using the model's lidar-to-image calibration and alpha-composited into all camera views and input frames before normalization. Each edit remains fixed across the input queue. The insertion procedure applies visibility checks but no map or interaction constraints.

Each predicted plan contains six waypoints at 0.5 s intervals over a 3 s horizon. Future traffic participants are obtained from nuScenes annotations at the corresponding keyframes, without temporal interpolation. For a common post-hoc evaluation, each victim's plan and the future agent boxes are expressed in a common per-sample ego-centered frame at t_0 using the corresponding sensor-time ego poses. At each future keyframe, a collision is recorded when the predicted ego oriented polygon intersects any ground-truth agent polygon. RQ-specific sample eligibility, attack settings, manipulation ranges, and analysis procedures are described in Sections 5–7.

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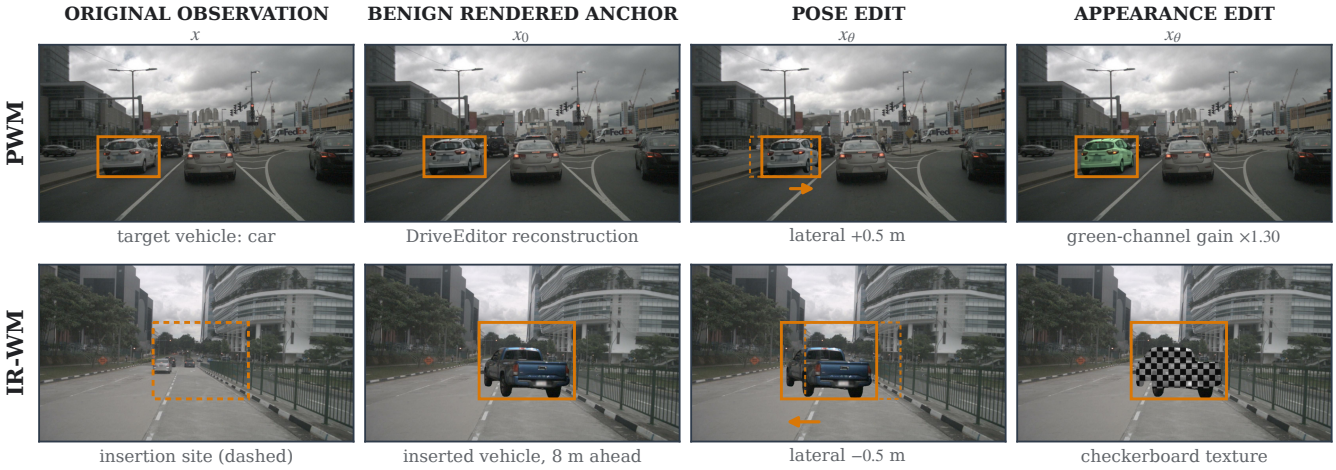


Fig. 2 Representative rendered inputs used in RQ2. From left to right, each row shows the original observation x, the benign rendered anchor x_0, and one pose and one appearance edit x_θ. PWM reconstructs an existing target vehicle, whereas IR-WM inserts a vehicle 8 m ahead. Orange boxes mark the manipulated vehicle, and the dashed box marks the IR-WM insertion site. One front-view frame is shown for readability; each edit is applied consistently across the victim's complete visual input.

Table 1 Representation-level attack results. Eligible samples have valid, collision-free clean plans. Rand. is the mean D-ASR of five support- and norm-matched random controls.

Surface	Eligible	Rel. size	CFR	D-ASR	Rand.
PWM–dyn.	96	2.300	7.3%	5.2%	5.4%
PWM–ctx.	96	1.482	24.0%	9.4%	0.0%
IR-WM	97	0.116	75.3%	42.3%	0.0%

5. Direct Representation Perturbations (RQ1)

5.1 Attack Setup

For PWM, we perturb the pre-LFQ representation `z_e` in the dynamic and context branches separately. The attacked past-token supports contain 308×13 and 448×13 latent elements, respectively. We run 80 zero-initialized projected sign-gradient steps with $\|\delta\|_\infty \leq 5.0$. Gradients through LFQ use a hard-forward, soft-backward straight-through estimator whose temperature is annealed from 1.0 to 0.05, and the final representation is hard-requantized before planning. For IR-WM, we perturb the fused `ref_bev` representation with 20 zero-initialized steps and a per-sample budget of 0.2 mean $|z|$, where mean $|z|$ is the mean absolute entry of the clean representation. Gradients are propagated through all five predicted future states. Neither attack uses random restarts.

We use CFR and D-ASR as defined in Section 3.3. Here, the reference is the clean plan, so N_s contains valid samples with collision-free clean plans. Five independently seeded random perturbations, matched to the optimized perturbation in support and ℓ_2 norm, serve as controls. Relative perturbation size is measured by $\|\delta\|_2/\|z\|_2$ on the attacked support.

5.2 Results

Table 1 reports the results for the three attacked surfaces. IR-WM shows the clearest representation-level vulnerability. The optimized perturbation reaches a D-ASR of 42.3%, while all five matched random controls remain at zero. Optimization therefore identifies safety-critical directions that are not obtained from arbitrary perturbations of the same support and norm.

PWM exhibits different behavior across its two representation channels. In the context branch, the optimized attack reaches 9.4% D-ASR against a zero random baseline, again indicating structured attack directions. In the dynamic branch, optimized and random perturbations produce nearly identical success rates. The vulnerability exposed in PWM is therefore concentrated in the context representation under the evaluated attack.

The relative-size column shows that the three surfaces are evaluated at different perturbation scales. The matched random controls provide the within-surface comparison between optimized and arbitrary directions. CFR is also consistently higher than D-ASR. Across the three surfaces, 48 of 103 collision flips fail the directed conditions, and 45 of those reduce terminal forward progress. We therefore use D-ASR as the primary success measure in the cross-level experiments.

RQ1: Optimized direct perturbations identify failure-inducing directions in IR-WM and PWM's context representation: their D-ASRs are 42.3% and 9.4%, respectively, while matched random controls remain at zero. PWM's dynamic branch shows no corresponding separation from the random controls.

6. Rendered Object Manipulation (RQ2)

RQ1 shows that direct changes to the internal representation can alter planning behavior. We now test whether visible changes to one vehicle can reproduce those failures through rendering, encoding, world-model rollout, and plan-

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Table 2 End-to-end outcomes of the rendered object manipulations. The last two columns report success counts over pair-safe samples.

Victim	Manipulation	Pair-safe	N_{flip}/N_s	N_{dir}/N_s
PWM	Pose + scale	4/5	0/4	0/4
PWM	Appearance	4/5	0/4	0/4
IR-WM	Pose + scale	24/24	0/24	0/24
IR-WM	Appearance	24/24	0/24	0/24

ning.

6.1 Rendered Object Manipulations

Each candidate changes one attribute of the benign rendered anchor x_0 shown in Fig. 2.

The pose-and-scale endpoints are shared across victims: lateral shifts of ± 0.5 m, longitudinal shifts of ± 1.0 m, yaw changes of $\pm 15^\circ$, and isotropic scales of 0.8 and 1.2. Each edit is held constant across the input frames; IR-WM projects the same 3D edit into all six camera views.

PWM additionally evaluates eight target-local photometric edits: brightness factors 1.25 and 0.78, contrast factors 1.30 and 0.72 about the target-mask mean, a 1.30 gain on each RGB channel, and a 50% blend with grayscale. Pixels outside the target mask remain unchanged. IR-WM evaluates five geometry-preserving vehicle textures: dark, light, checkerboard, high-contrast, and adversarial-texture variants. The two appearance parameterizations are therefore victim-specific. The larger endpoint directions used by the RQ3 joint counterfactual are defined separately in Section 7.2.

Every candidate is passed through the complete victim pipeline. The prescribed sets are fully enumerated, giving 16 edited planner evaluations per PWM object and 13 per IR-WM object, in addition to the benign reference. We retain the strongest candidate in each manipulation family, prioritizing directed success, then collision occurrence, and then the attack objective. This lexicographic selection preserves any success present in the evaluated set rather than averaging across conditions.

Figure 2 also makes the paired reference explicit: x_0 already contains the PWM reconstruction or the IR-WM insertion before any evaluated attribute edit is applied. We therefore define a sample as pair-safe when both the original observation x and the benign rendered anchor x_0 produce collision-free plans, and evaluate each edited input relative to x_0. Pair-safe samples form the denominator N_s in Eq. (3). The edited vehicle is excluded from the attack objective, collision target set, and D-ASR target set, so success requires the edit to redirect the ego plan into another traffic participant.

6.2 End-to-End Results

For PWM, one benign reconstruction already produces an unsafe plan, leaving four pair-safe samples. None of the evaluated pose, scale, or appearance conditions induces a collision on those samples. All 24 IR-WM samples are pair-safe, and none of its rendered conditions induces a collision. CFR and D-ASR are therefore zero for every victim and manipulation family in **Table 2**.

6.3 Paired Representation Changes

We next compare each rendered attack with a direct representation attack that starts from the same benign representation $z_0 = E(x_0)$. Both changes are measured on the same attacked support. For each sample, $\theta^\star_{\text{obj}}$ is selected across the union of pose-and-scale and appearance candidates by the same lexicographic rule, and $d_{\text{obj}} = E(x_{\theta^\star_{\text{obj}}}) - z_0$. Relative size is the ℓ_2 norm of the displacement divided by the norm of z_0 on the attacked support.

For IR-WM, the object-induced and direct changes have similar overall scale: their median relative sizes over the 24 paired samples are 0.121 and 0.114. The paired direct attack reaches D-ASR on 14 samples, whereas the strongest rendered object manipulation reaches D-ASR on none. The paired outcomes consist of 14 latent-only successes and 10 cases in which neither attack succeeds. These 14 latent-only cases form the main analysis set in RQ3.

Within that set, the median object-induced change remains close to the successful direct perturbation, at 0.095 versus 0.114. The cross-level difference is therefore visible even when both attacks start from the same representation and produce changes of similar size. PWM context has no paired direct success among its four pair-safe samples, so it does not provide the same success-versus-failure comparison.

RQ2: None of the enumerated pose, scale, or appearance edits produces a collision flip on the pair-safe samples. In the paired IR-WM cases, object-induced changes are similar in magnitude to successful direct perturbations, but no edited plan reaches directed success.

7. Directional Analysis of the Gap (RQ3)

RQ2 shows that object edits can produce a sizeable representation change without reproducing a successful direct attack. We now test whether the two attack levels reach different directions in the representation space.

7.1 Alignment of Object-Induced Directions

We analyze the 14 IR-WM samples for which the paired direct attack succeeds. For sample j, the eight pose-and-scale and five appearance displacements form the unnormalized matrix $D_j = [d_{j,1}, \ldots, d_{j,13}]$. We compute its thin singular value decomposition $D_j = U_j \Sigma_j V_j^\top$ and use the nonzero left singular vectors as an orthonormal basis of $S_j = S(o_j)$. Numerical rank uses the tolerance $\max(D, 13)\epsilon_{\text{mach}}\sigma_1$, where the attacked representation has ambient dimension $D = 10{,}240{,}000$. The energy dimension

$$k_{90} = \min\left\{ k : \frac{\sum_{q=1}^{k} \sigma_q^2}{\sum_q \sigma_q^2} \geq 0.9 \right\} \tag{11}$$

shows that the displacement sets contain several distinct directions: at the median, five pose-and-scale components and three appearance components are needed to explain 90% of

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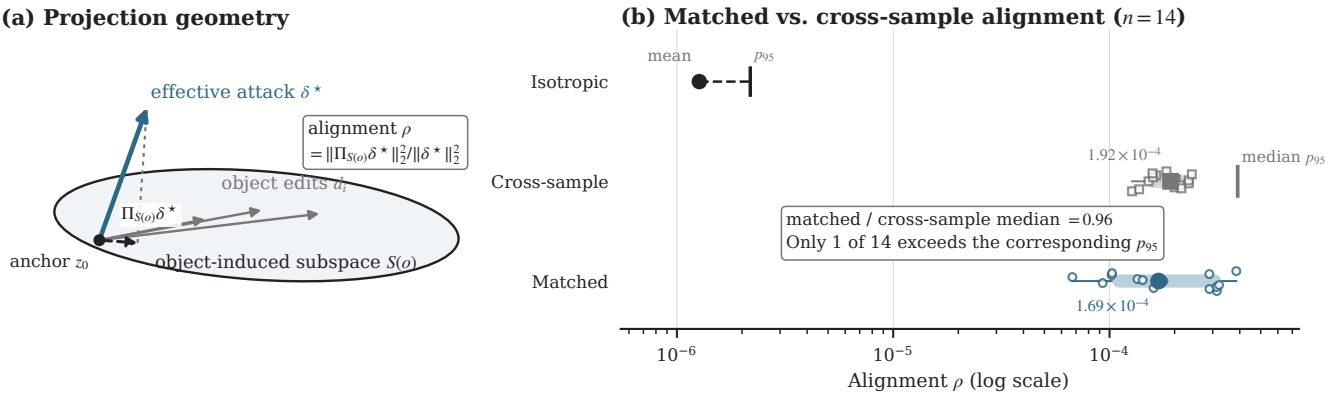


Fig. 3 Geometric comparison between effective IR-WM attacks and object-induced directions. (a) Projection energy measures the part of a successful direct attack captured by the object-induced subspace. (b) Each matched projection is compared with the 23 other-sample attacks for the same subspace. Both the matched and cross-sample projections exceed the rank-matched isotropic baseline.

their energy.

Across the 14 samples, the median alignment in Eq. (9) is 1.69×10^{-4} (**Fig. 3**(a)), so the object-induced span captures only 0.0169% of the successful attack energy. The best individual edit is also nearly orthogonal: the maximum signed cosine in Eq. (10) has a median of 0.0064. The family-specific medians show the same pattern: 1.13×10^{-4} for pose and scale and 5.74×10^{-5} for appearance.

To distinguish paired correspondence from shared representation structure, we define

$$\rho_{j,k} = \frac{\|\Pi_{S_j}\delta_k^\star\|_2^2}{\|\delta_k^\star\|_2^2}. \tag{12}$$

The matched value is $\rho_{j,j}$. For each of the 14 successful matched attacks, the 23 values with $k \neq j$ use attack directions optimized for the other paired samples and form that subspace's cross-sample reference distribution. Its 95th percentile is therefore computed separately for each j.

Under the isotropic model, ρ follows $\mathrm{Beta}(r/2, (D-r)/2)$; its 95th percentile provides the rank-matched isotropic threshold. The matched projection is 132.8 times the isotropic expectation, but the cross-sample projections are also strongly above isotropy, at 150.9 times the same reference at their median (Fig. 3(b)). The median of the per-sample matched-to-cross-sample ratios is 0.96, the median cross-sample percentile is 0.457, and only one of the 14 matched values exceeds its own cross-sample 95th percentile. The isotropic excess therefore reflects anisotropic structure shared across samples, while the correct sample pairing contributes no additional alignment.

7.2 Joint Search within the Bounded Object-Induced Region

A successful direction may combine several attributes. Separately from the small-step basis used above, we construct 13 box-edge endpoint displacements using lateral shifts of ±2.0 m, longitudinal shifts of ±4.0 m, yaw changes of $\pm60^\circ$, scales of 0.8 and 1.2, and the same five textures. We jointly optimize their coefficients in $\mathcal{R}_{\mathrm{edge}}(o)$ from Eq. (7), allowing cross-family combinations while selecting at most one side of each signed pair and one texture. The coefficient search uses 120 zero-initialized projected sign-gradient steps with step size 2.5/120, one deterministic run, and the best collision-surrogate iterate. It is a latent linear counterfactual, not an end-to-end jointly rendered attack.

The joint search produces no collision flip and no directed success on any of the 24 paired samples. Its median realized ℓ_∞ displacement is 24.7 times the direct-attack budget, while direct PGD using the same support and objective but unconstrained to the object-induced region reaches D-ASR on 14 of the same 24 benign representations. The search nevertheless changes the plan, with a median waypoint ADE of 0.122 m, but the minimum attacked clearance remains 1.186 m. Increasing the displacement within the evaluated object-induced region does not recover the effective attack directions.

RQ3: The paired IR-WM analysis indicates a directional mismatch. Object-induced subspaces capture a median fraction of 1.69×10^{-4} of the successful attack energy; matched pairs show no alignment advantage over cross-sample pairs, and the bounded joint search produces no directed success.

8. Discussion

Our results distinguish representation-space vulnerability from external reachability. Direct perturbations test whether safety-critical directions exist near a representation; object-level attacks test whether observable variables can reach them through the encoder. The former is a sensitivity question and the latter a reachability question. Representation-level attacks are therefore diagnostic probes; external exploitability additionally depends on the directions generated by observable control variables.

The object-induced span provides a local controllability view. For continuous pose and scale parameters a, finite differences approximate directions in the range $\mathrm{Im}(J_a)$ of the encoder Jacobian $J_a = \partial E(x(a))/\partial a$, while appearance variants provide secant directions. Thus, $S(o)$ is an em-

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pirical linearization of the evaluated edits rather than the complete set of representations reachable through arbitrary object manipulation. Projection measures how much of a paired effective direction is captured, while the bounded joint search tests whether another combination of the observed directions can produce a failure.

Although the measured projections exceed a rank-matched isotropic baseline, other-sample attack directions show the same excess. An isotropic baseline alone would therefore confound shared anisotropic structure with paired reachability. The cross-sample control directly tests whether the correct object–attack pairing adds alignment; in the paired IR-WM set, it does not. Increasing coefficients or combining pose and appearance consequently explores a larger but still misaligned region.

This perspective suggests a way to screen other attack surfaces. A direct attack first provides a local target $\delta^\star$. Candidate external variables can then be evaluated by the alignment of their induced representation directions with $\delta^\star$, rather than only by change magnitude. Relevant candidates include coordinated multi-object motion, interaction patterns, occlusion or visibility changes, and scene- or conditioning-level inputs that affect the rollout more directly. Robustness evaluation should therefore characterize both internal sensitivity and environmental controllability. This two-level view makes the representation-to-object gap a guide for locating the next externally relevant attack surface.

9. Conclusion

We examined whether representation-level vulnerabilities in world-model-based planners can be reached through visible-object manipulation. Direct perturbations identified failure-inducing directions in IR-WM and PWM's context representation, whereas none of the tested pose, scale, or appearance edits caused a collision flip on the pair-safe samples. In paired IR-WM cases, object-induced changes had similar magnitude to successful direct perturbations but no matched-sample alignment advantage; the bounded joint counterfactual also produced no directed success. The tested edits therefore exhibit a directional gap: changing the representation is insufficient unless the change reaches planning-relevant directions. Future work should examine externally controllable variables with stronger alignment.

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